Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/19/2026
PI Name:	Brainard, George C. Ph.D.		
Project Title:	Testing Solid State Lighting Countermeasures to Improve Circadian Adaptation, Sleep, and Performance During High Fidelity Analog and Flight Studies for the International Space Station		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sleep**:Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	george.brainard@jefferson.edu	**Fax:**	FY 215-923-7588
PI Organization Type:	UNIVERSITY	**Phone:**	215-955-7644
Organization Name:	Thomas Jefferson University		
PI Address 1:	Light Research Program		
PI Address 2:	1025 Walnut St., Room 507		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19107-5083	**Congressional District:**	1
Comments:			
Project Type:		**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	12/01/2014	**End Date:**	06/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	Flight Definition NOTE: End date changed to 06/30/2025 per NSSC information (Ed., 9/17/24) NOTE: End date changed to 08/31/2024 per L. Juliette/JSC (Ed., 4/10/23) NOTE: End date changed to 08/31/2023 per NSSC information (Ed., 10/13/22) NOTE: End date changed to 08/31/2022 per NSSC information (Ed., 8/31/21) NOTE: End date changed to 08/31/2021 per NSSC information (Ed., 2/3/21) NOTE: End date changed to 12/31/2020 per D. Risin/HRP/NSSC information (Ed., 09/14/2020) NOTE: End date changed to 09/30/2020 per NSSC information (Ed., 04/27/2020) NOTE: End date changed to 03/31/2020 per NSSC information (Ed., 9/3/19) NOTE: End date changed to 11/30/2019 per NSSC information (Ed., 10/11/18) NOTE: End date changed to 11/30/2018 per NSSC information (Ed., 12/13/17) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance		

	(Ed., 1/17/17)
Key Personnel Changes/Previous PI:	December 2019 - Smith Johnston, MD, retired from NASA but has stayed active on this project.
COI Name (Institution):	Barger, Laura Ph.D. (Brigham and Women's Hospital/Harvard Med Ctr) Clark, Toni B.S. (NASA Johnson Space Center) Czeisler, Charles M.D., Ph.D. (Brigham and Women's Hospital/Harvard Medical Center) Johnston, Smith M.D. (NASA Johnson Space Center (Retired 12/19, but still involved)) Moomaw, Ronald O.D. (NASA Johnson Space Center) Lockley, Steven Ph.D. (Co-PI: Brigham and Women's Hospital) Hanifin, John Ph.D. (Thomas Jefferson University) Rahman, Shadab Ph.D. (Brigham and Women's Hospital) St Hilaire, Melissa Ph.D. (Brigham and Women's Hospital)
Grant/Contract No.:	NNX15AC14G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This research addresses the NASA Research Announcement (NRA) NNJ13ZSA002N-BMED: Behavioral Health and Human Performance: "Evaluation of the Neurobehavioral Effects of a Dynamic Lighting System on the ISS." This NRA solicited both "Ground Based and Flight-Definition" research with the specific instructions that the "ground study serves as a precursor to the flight study, therefore the ground study should take place in an analog with high fidelity to the ISS. The SSLAs should be studied in a high fidelity ground analog environment, then implemented on ISS to evaluate individual crewmember outcomes related to circadian physiology, sleep, behavioral health and performance using sensitive and validated measures that are feasible in the spaceflight environment." Originally, the International Space Station (ISS) used General Luminaire Assemblies (GLAs) that housed fluorescent lamps for illuminating the astronauts' working and living environments. NASA determined that, beginning in 2016, the GLAs would be replaced with Solid-State Light Assemblies (SSLAs) containing Light Emitting Diodes (LEDs). Engineers at Kennedy Space Center developed a prototype Solid-State Lighting Assembly (SSLA) that was successfully installed onboard the ISS during ISS Expedition 18. The Principal Investigator and Co-Principal Investigator of the intended research worked with engineers, scientists, and managers from Johnson Space Center (JSC) to revise the SSLA specifications so that the new lighting units would have dual capacity to: 1) provide illumination for crew members' working and living quarters, and 2) serve as a lighting countermeasure for crewmembers' circadian and sleep disruption. NASA ordered and received for a set of SSLAs intended to have this dual capacity. This research is comprised of a multidisciplinary collaboration between Thomas Jefferson University, Brigham and Women's Hospital, and JSC to complete a ground-based study in a high fidelity analog of the crew sleeping quarters and daily living environment of the ISS. Specifically, standardized psychometric, physiological, and neurobehavioral measures are testing the efficacy of light from the SSLAs to improve vision, circadian regulation, sleep, and performance in healthy astronaut-aged subjects. In addition, the initial SSLA was installed on ISS in 2016. Since then, a total of 82 SSLAs have been installed on ISS, bringing the total retrofit to 96% replacement of GLAs on the US portion of ISS. Since the onset of the SSLA retrofit, the investigators started the inflight ISS study to assess the acceptability, use, and impact of deployment of a dynamic lighting schedule aboard the ISS during operational flight missions on astronaut vision, sleep, alertness, circadian rhythms, and general well-being. Sleep, performance, and circadian rhythm data will be compared to those collected by their team and others during previous flight missions aboard ISS, in addition to surveillance of medical and psychological health in collaboration with mission flight surgeons. This project will generate quantitative data and knowledge for the benefit of crew health, habitability, environment, and human factors in the design of future human spaceflight vehicles and habitats. The project also will provide guidance for flight surgeons, flight psychologists, and astronauts to help optimize sleep and circadian regulation during space exploration missions. This research was designed to address NASA's Program Requirements Document (PRD) Risk: "Risk of Performance Errors due to Fatigue Resulting from Sleep Loss, Circadian Desynchronization, Extended Wakefulness and Work Overload" and Integrated Research Plan (IRP) Gap Sleep5: "We need to identify environmental specifications and operational regimens for using light to prevent and mitigate health and performance decrements due to sleep, circadian, and neurobehavioral disruption, for flight, surface, and ground crews, during all phases of spaceflight operations." The results of this research also specifically address other high priority risks of the Human Factors and Behavioral Performance Element, including the Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders, and the Risk of an Incompatible Habitat Design. Appropriately designed lighting systems will serve as a countermeasure to mitigate such risks in future Exploration missions. Importantly, this work will lead to advances in new lighting systems for civilian applications in work places and homes. As of November 2021, the Human Research Program risks and gaps have since been revised per more recent IRPs; see above for the current noted Risks and Gaps and the Human Research Roadmap: https://
Rationale for HRP Directed Research:	
	The sleep deficits experienced by astronauts during spaceflight along with risk of incompatible habitat design can be considered threats to the success of space missions (NASA Human Research Program Integrated Risk Plan, 2022). The resulting physiological and behavioral changes caused by sleep and circadian disruption can lead to diminished alertness, cognitive ability and psychomotor performance (Dijk et al., Amer. J. Physiol., 2001; Human Health and Performance Risks of Space Exploration Missions. McPhee and Charles, eds., 2010). As a measure to counteract sleep disruptions, crew members report using sleep promoting drugs: 71% on space shuttle flights and 75% during ISS expeditions (Barger et al., Lancet Neurology, 2014; Flynn-Evans et al., 2016 npj Microgravity). A significant portion of the global population suffers from chronic sleep loss and/or circadian-related disorders. Evidence for disease occurring due to a disruption of circadian homeostasis has mounted significantly in the past several years. In the United States, nearly 22 million Americans do shift work that interferes with a biologically healthy nocturnal sleep cycle (US Bureau of Labor Statistics, 2007). It has been shown that shift workers are more likely to suffer from a wide variety of ailments, including cardiovascular disease, metabolic disorders, gastrointestinal distress, and cognitive and emotional problems.

Research Impact/Earth Benefits:	Development of an in-flight lighting countermeasure that helps resolve circadian and sleep disruption in astronauts is likely to help optimize the use of light therapy for patient populations with affective, circadian and sleep disorders. References Dijk DJ, Neri DF, Wyatt JK, Ronda JM, Riel E, A. R-D, Hughes RJ, Elliott AR, Prisk GK, West JB and Czeisler CA (2001) Sleep, performance, circadian rhythms, and light-dark cycles during two space shuttle flights. Am J Physiol 281:R1647-R1664. McPhee J and Charles J, eds. 2010. Human Health and Performance Risks of Space Exploration Missions: Evidence Reviewed by the NASA Human Research Program. NASA SP-2009-3405 edition. Barger LK, Flynn-Evans EE, Kubey A, Walsh L, Ronda JM, Wang W, Wright KP and Czeisler CA (2014) Prevalence of sleep deficiency and use of hypnotic drugs in astronauts before, during, and after spaceflight: an observational study. Lancet Neurology 13:904-912. Flynn-Evans EE, Barger LK, Kubey AA, Sullivan JP and Czeisler CA (2016) Circadian misalignment affects sleep and medication use before and during spaceflight. npj Microgravity 2:15019; doi:15010.11038/npjmgrav.12015.15019.
Task Progress:	Three ground analog studies and an International Space Station (ISS) flight study were completed successfully in this project. In the ground analog studies, a series of visual performance tests (VPTs) were performed under 3 pre-selected lighting conditions, using adapted Solid State Lighting Modules-Research (SSLM-R) units inside a high-fidelity replica of the ISS crew quarters. For the ISS flight study, 37 sessions of light measurements consisting of spectrometer data of General Luminaire Assemblies (GLAs) and newly installed Solid-State Lighting Assemblies (SSLAs), were taken between December 2016 and September 2019. [Ed. Note: For preliminary results of the study, please see the FY23 NASA Task Book report: https:// .] An augmentation of the project was approved by the NASA Human Factors and Behavioral Performance (HFBP) Element and Thomas Jefferson University (TJU) received a NASA supplement contract from the NASA Human Research Program in April 2024. In the current study, we have completed initial assessments of the stability and utility of the new SSLA lighting system Pre-Sleep setting and have measured its physiological impact on circadian phase, sleep, and performance. Exposure to both the original and newly developed Pre-Sleep settings resulted in significantly greater production of melatonin compared to the General Vision setting. Compared to the General Vision setting, color vision was significantly compromised under both Pre-Sleep settings. Complete results of these studies will be detailed in future publications. Based on analysis of the analog and flight data, the study team strongly recommends continuation of the flight study now that the retrofit of ISS with SSLAs are nearly complete (99%). These data can inform the lighting technologies and designs for the Artemis Mission, Gateway Mission, and other future exploration missions.
Bibliography Type:	Description: (Last Updated: 01/19/2026)
Abstracts for Journals and Proceedings	Brainard G, Rahman S, Clark T, St Hilaire M, Garcia D, Norwood K, Kemp J, Hanifin J, Warfield B, Yoo T, Maida J, Johnston S, Moomaw R, Barger L, Czeisler C, Lockley S. "Testing solid state lighting countermeasures to improve circadian adaptation, sleep, and performance during high fidelity analog and flight studies for the International Space Station." 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Brainard G, Hanifin J, Serruya M, Warfield B, Kemp J, Yoo T, Rahman S, Clark T, St Hilaire M, Soler R, Garcia D, Norwood K, Maida J, Johnston S, Moomaw R, Barger L, Czeisler C, Lockley S. "Testing solid state lighting countermeasures to improve circadian adaptation, sleep, and performance during high fidelity analog and flight studies for the International Space Station." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Hanifin J, Serruya M, Warfield B, Kemp J, Yoo T, Ayers M, Soler R, Lockley S, Brainard G. "Solid state lighting countermeasures to improve color vision and melatonin onset during a high fidelity analog study for the International Space Station." 35th Annual Meeting of the Society for Light Treatment and Biological Rhythms, Prague, Czech Republic, June 20-June 22, 2024. Abstracts. 35th Annual Meeting of the Society for Light Treatment and Biological Rhythms, Prague, Czech Republic, June 20-June 22, 2024. , Jun-2024
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Abstracts for Journals and Proceedings	Brainard G, Hanifin J, Serruya M, Warfield B, Kemp J, Yoo T, Soler R, Rahman S, St Hilaire M, Clark T, Norwood K, Garcia D, Maida J, Johnston S, Moomaw R, Czeisler C, Barger L, Lockley S. "Ground and flight studies testing the visual and physiological effects of tunable LED lighting for the International Space Station." 36th Annual Meeting of the Society for Light, Rhythms, and Circadian Health (SLRCH), Boston, Massachusetts, June 14-16, 2025. Abstracts. 36th Annual Meeting of the Society for Light, Rhythms, and Circadian Health (SLRCH), Boston, Massachusetts, June 14-16, 2025. , Jun-2025
Articles in Peer-reviewed Journals	Grant LK, Kent BA, Rahman SA, St Hilaire MA, Kirkley CL, Gregory KB, Clark T, Hanifin JP, Barger LK, Czeisler CA, Brainard GC, Lockley SW, Flynn-Evans EE. "The effect of a dynamic lighting schedule on neurobehavioral performance during a 45-day simulated space mission." Sleep Adv. 2024 May 30;5(1):zpae032. https://doi.org/10.1093/sleepadvances/zpae032 ; PMID: 38903700 ; PMCID: PMC11187988 , May-2024